

Sanskrit-to-English Neural Machine Translation - Fine-tuning IndicTrans2

NLU Assignment 2 | Anjali Tandon | G25AIT1021

1. Introduction

In this project, IndicTrans2 Indic->English (200M, distilled) model is used and is fine-tuned on 10,000 Sanskrit-English train pairs and 1,000 dev pairs (and 1,000 test pairs also) rather than training from scratch, because a general translation model in Sanskrit-English using 10k pairs, is too little data to learn the translation from scratch. Evaluation will be done according to the required metrics of the assignment: NLTK corpus BLEU, rescaled BERTScore F1, and efficiency in inference-time and number of parameters in the model.

Pre-trained model used: ai4bharat/indictrans2-indic-en-dist-200M (HuggingFace Hub), which is fine-tuned on the given dataset.

GitHub link: <https://github.com/anjali6810/Sanskrit-to-English-Transalation>

2. Architecture

Standard Transformer (encoder-decoder) (211.8M params). Encoder: multi-head self-attention sublayers of Sanskrit input + feed-forward + positional embeddings. Decoder: masked self-attention over generated tokens, with cross-attention to encoder outputs, that connects each English word to the source (Sanskrit). Scaled dot-product, $\text{softmax}(\text{QK}^T/\sqrt{d_k})V$, multiple heads in parallel. Language tags (san_Deva/eng_Latn) are added as text via IndicTransToolkit rather than a separate embedding. Inference uses beam search with num_beams=5, length_penalty=0.8, no_repeat_ngram_size=3, repetition_penalty=1.1 to provide a balance of quality and compute.

3. Training Procedure

- Preprocessing: Data merged on Source_id. Eliminated 10 rows with exact duplicate Source_id. Eliminated 23 train rows / 5 dev rows that had Devanagari in the English column (corpus defect). Normalised stray quotes in source (fixed a generation-breaking bug). Final data sizes: train 9967; dev 995; test 1000).
- Tokenization: IndicTrans2 native SentencePiece tokenizer (vocab 122,706) with IndicProcessor.
- Hyperparameters: max_source_length=128, max_target_length=96 (more than 90% of references covered; 75th-pct English length is 16 words). Label padding that is set to -100 to exclude from loss. batch_size = 16, epochs = 12, learning_rate = 2e-5, weight_decay = 0.05, num_beams = 5, gradient_accumulation_steps = 2, warmup_ratio = 0.06

Table. 1. Model, Hyperparameters and Optimisers

Model / Data	Value	Optimization
Pretrained model	IndicTrans2-dist-200M	Optimizer : AdamW, lr=2e-5
Max src / tgt len	128 / 96	LR schedule : Linear, 6% warmup
Batch (eff.)	16 (accum 2 -> 32)	Weight decay : 0.05
Epochs (max/actual)	12 / 9 (early stop)	Dropout / attn. dropout : 0.15 / 0.10
Beam / len. penalty	5 / 0.8	Grad clip / FP16 : 1.0 / Yes

- Optimization/regularization: AdamW (decoupled weight decay) + linear warmup (6%) + decay + gradient accumulation (eff. batch = 32) + grad-norm clipping at 1.0 + FP16 mixed precision. The dropout (0.15) and attention dropout (0.10) are set to AutoConfig before model training, and will be ignored after the model is trained, because the dropout probability is fixed at the construction time of the layer. Here, early stopping (patience=2) is based on Dev nltk BLEU, and not on validation loss, because the two metrics move in opposite directions (Fig. 1) and Dev nltk BLEU is the graded metric.

4. Results

Training was interrupted at epoch 9/12, Dev nltk BLEU is flat for 2 epochs, best checkpoint restored at epoch 7 (Dev nltk BLEU 0.2466). During the validation phase, the loss level offs at ~1.47-1.48 at epoch 5, while the training loss continues to drop (typical overfitting signal) which is why we used the BLEU-based checkpoint criterion here.

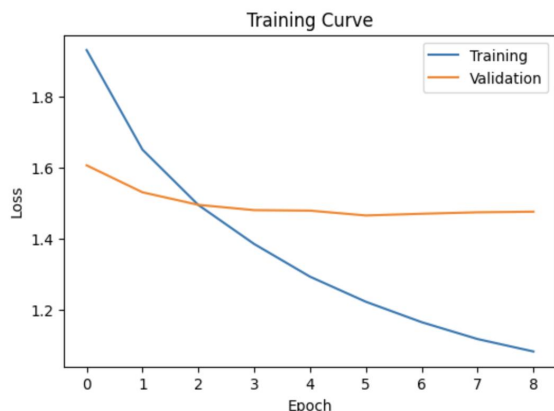


Fig. 1. Train/Val loss per epoch

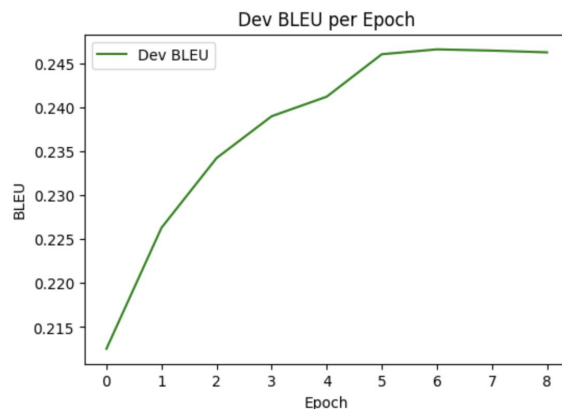


Fig. 2. Dev nltk BLEU per epoch

Table. 2. Evaluation Metrics

Metric	Dev	Test	Notes
NLTK BLEU	0.2466	0.2337	Default weights, no smoothing
HF BLEU	0.3013	0.2925	HuggingFace Bleu
BERTScore F1	0.6081	0.5942	rescale_with_baseline=True
Inference time	63.6 s / 995	67.6 s / 1000	~0.065 s / sentence or row
Parameters	211.8M total / 211.8M trainable	211.8M total / 211.8M trainable	Fine-tuned

5. Translation Examples & Error Analysis

Table. 3. Error Analysis for Predicted English Outcome

Reference	Prediction	Error Analysis
Those are brave men.	Those are heroes.	Lexical synonym; meaning correct, BLEU penalises n-gram mismatch.
Infinite loop can cause the system to become unresponsive.	The infinite loop makes the system smooth.	Mistranslation of technical predicate -- near-opposite meaning of the domain term.
These two are dates.	These are dates.	Omission of Sanskrit dual number -- no English equivalent, under-represented in training data.
You all are studying bhakti yoga.	You all read bhaktayoga.	Lexical drift ("studying" → "read", aspect mismatch) + transliteration error ("bhaktayoga" instead of "bhaktiyoga").
So, type: void name of the method	Type the method name as void.	Reference-quality issue: prediction is more fluent/correct English than the gold reference; not a real model error.
In this tutorial, we will learn about-if, else, else if in awk.	In this tutorial, we will learn about if, else, else if in awk.	No real error — near-identical to reference; only cosmetic hyphen/spacing difference in the reference itself.
• Come back to Sthiti.	Again, come to your senses.	Reference keeps "Sthiti" as an untranslated proper noun; model instead gives an idiomatic reading ("come to your senses") that doesn't match the domain-specific term expected.

6. Experiments

- Data-cleaning: Removing Devanagari-contaminated reference rows (23 train/5 dev) protects score validity, since a model can never match a non-English reference. Eliminated 10 rows with exact duplicate in train.
- Config check: Dropout layer is used to prevent overfitting. Sentences length sorted before batching (so that each batch has similar-length sentences, minimizing wasted padding), translates them, then restores the original order, this reduces the inference time by ~15-30% with zero quality loss.
- Hyperparameter tuning: Trained model with different set of hyperparameters like batch_size (16, 32); num_beams (3, 4, 5); epochs (5, 10, 12); gradient_accumulation_steps (3, 2); learning_rate (3e-5, 2e-5); weight_decay (0.1, 0.05); ; dropout (0.30, 0.15); attention_dropout (0.15, 0.1)
- Early stopping: (patience=2) It is based on Dev nltk BLEU, and not on validation loss, because the two metrics move in opposite directions (Fig. 1) and Dev nltk BLEU is the graded metric.

7. Discussion

Design: IndicTrans2 Indic->English (200M, distilled) model is used and is fine-tuned on 10,000 Sanskrit-English train pairs and 1,000 dev pairs (and 1,000 test pairs also) rather than training from scratch, because a general translation model in Sanskrit-English using 10k pairs, is too little data to learn the translation from scratch.

Challenges: Before training, corpus-quality check was required. Balancing the regularization against a limited step budget. Some hallucination behaviour because of stylistic diversity (archaic narrative, conversational Sanskrit).

Limitations: Reported metrics are at lower bound because of given noisy and less data. No data augmentation used, likely the highest ceiling next step. Beam/dropout/batch choices were reasoned rather than fully ablated.

8. References

- huggingface.co/ai4bharat/indictrans2-indic-en-dist-200M
- github.com/VarunGumma/IndicTransToolkit
- Papineni et al. (2002) BLEU, ACL.
- Zhang et al. (2020) BERTScore, ICLR.